

LAB EXPERIMENT RECORD

SET 9 – Online Learning Activity

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Date	03/10/2026
Day	Saturday

AIM

To analyze online learning activity using R programming by creating a histogram, boxplot, scatter plot, and monthly quiz score trend, and to develop an interactive Tableau dashboard..

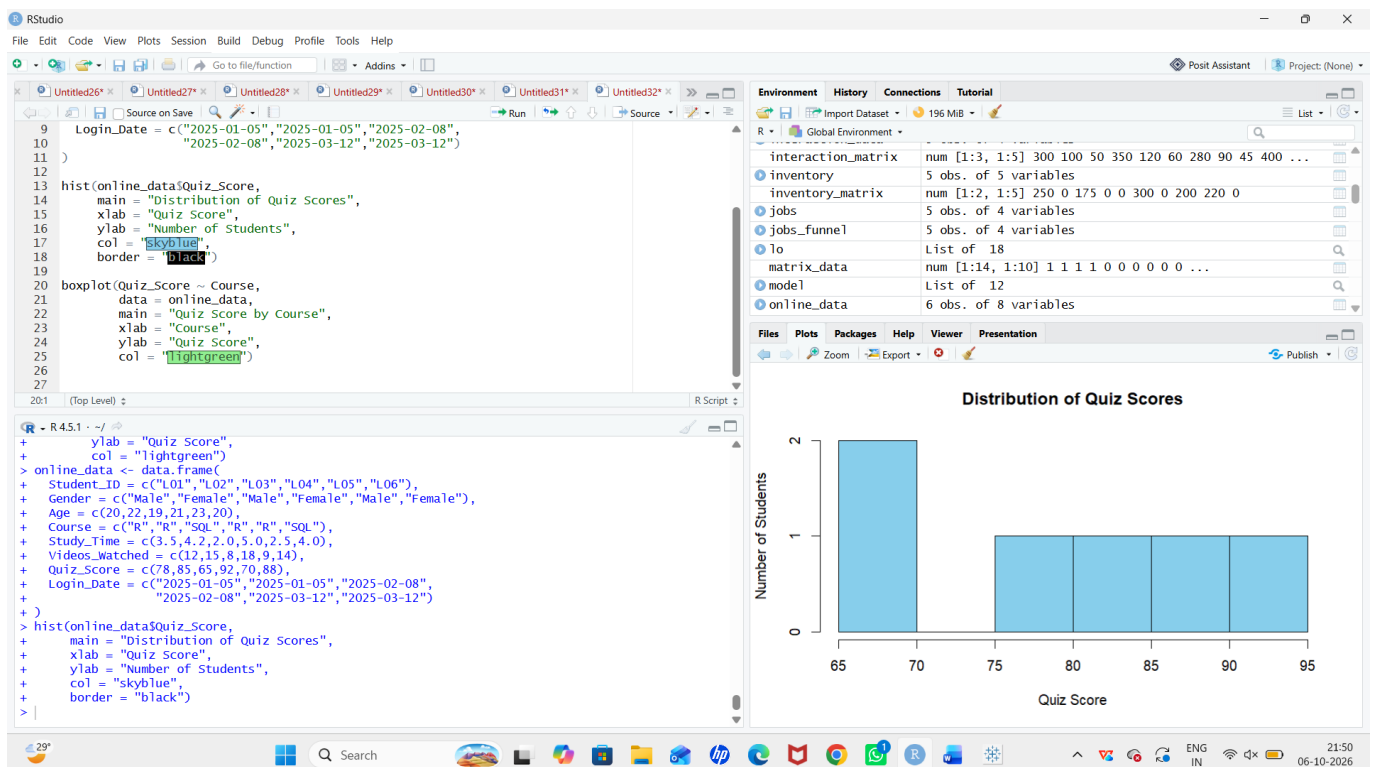
Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

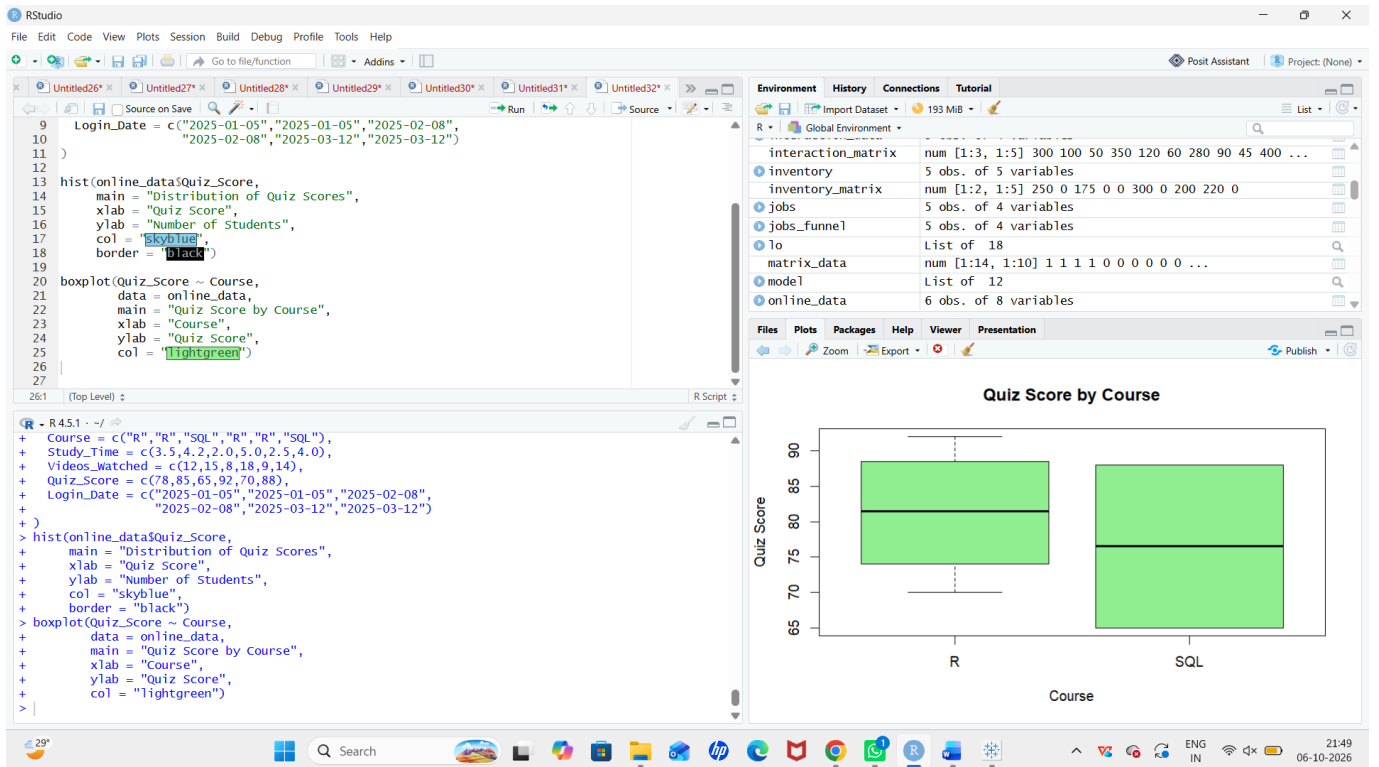
1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of

```
online_data <- data.frame(  
  Student_ID = c("L01","L02","L03","L04","L05","L06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Age = c(20,22,19,21,23,20),  
  Course = c("R","R","SQL","R","R","SQL"),  
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),  
  Videos_Watched = c(12,15,8,18,9,14),  
  Quiz_Score = c(78,85,65,92,70,88),  
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",  
    "2025-02-08","2025-03-12","2025-03-12")  
)
```

```
hist(online_data$Quiz_Score,  
  main = "Distribution of Quiz Scores",  
  xlab = "Quiz Score",  
  ylab = "Number of Students",  
  col = "skyblue",  
  border = "black")
```

```
boxplot(Quiz_Score ~ Course,  
  data = online_data,  
  main = "Quiz Score by Course",  
  xlab = "Course",  
  ylab = "Quiz Score",  
  col = "lightgreen")
```



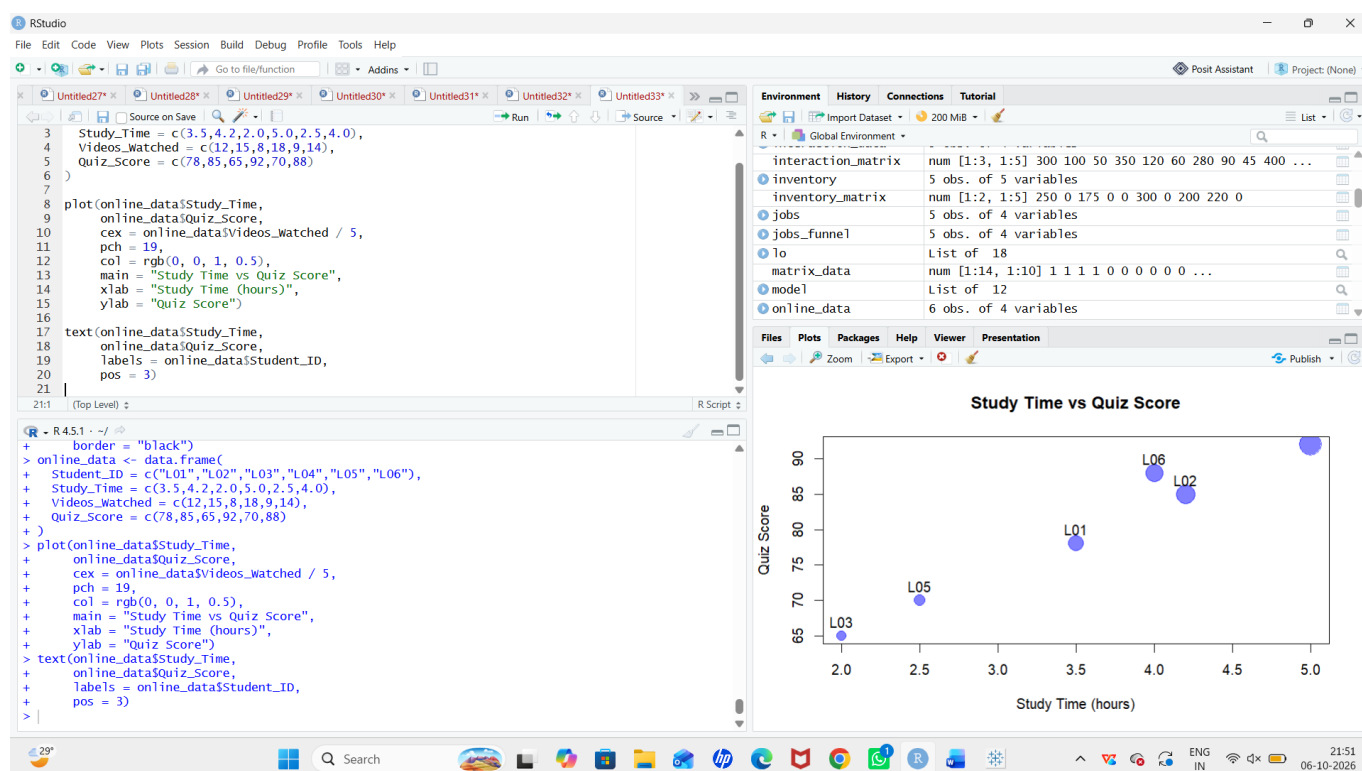


2. Create a scatter plot using R programming between Study_Time and Quiz_Score.

```
online_data <- data.frame(  
  Student_ID = c("L01","L02","L03","L04","L05","L06"),  
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),  
  Videos_Watched = c(12,15,8,18,9,14),  
  Quiz_Score = c(78,85,65,92,70,88)  
)
```

```
plot(online_data$Study_Time,  
     online_data$Quiz_Score,  
     cex = online_data$Videos_Watched / 5,  
     pch = 19,  
     col = rgb(0, 0, 1, 0.5),  
     main = "Study Time vs Quiz Score",  
     xlab = "Study Time (hours)",  
     ylab = "Quiz Score")
```

```
text(online_data$Study_Time,  
     online_data$Quiz_Score,  
     labels = online_data$Student_ID,  
     pos = 3)
```



3Using R programming, convert Login_Date into Date format..

```
online_data <- data.frame(
  Login_Date = c("2025-01-05", "2025-01-05", "2025-02-08",
                 "2025-02-08", "2025-03-12", "2025-03-12"),
  Quiz_Score = c(78, 85, 65, 92, 70, 88)
)

online_data$Login_Date <- as.Date(online_data$Login_Date)

online_data$Month <- format(online_data$Login_Date, "%Y-%m")

monthly_avg <- aggregate(Quiz_Score ~ Month,
                          data = online_data,
                          FUN = mean)

print(monthly_avg)

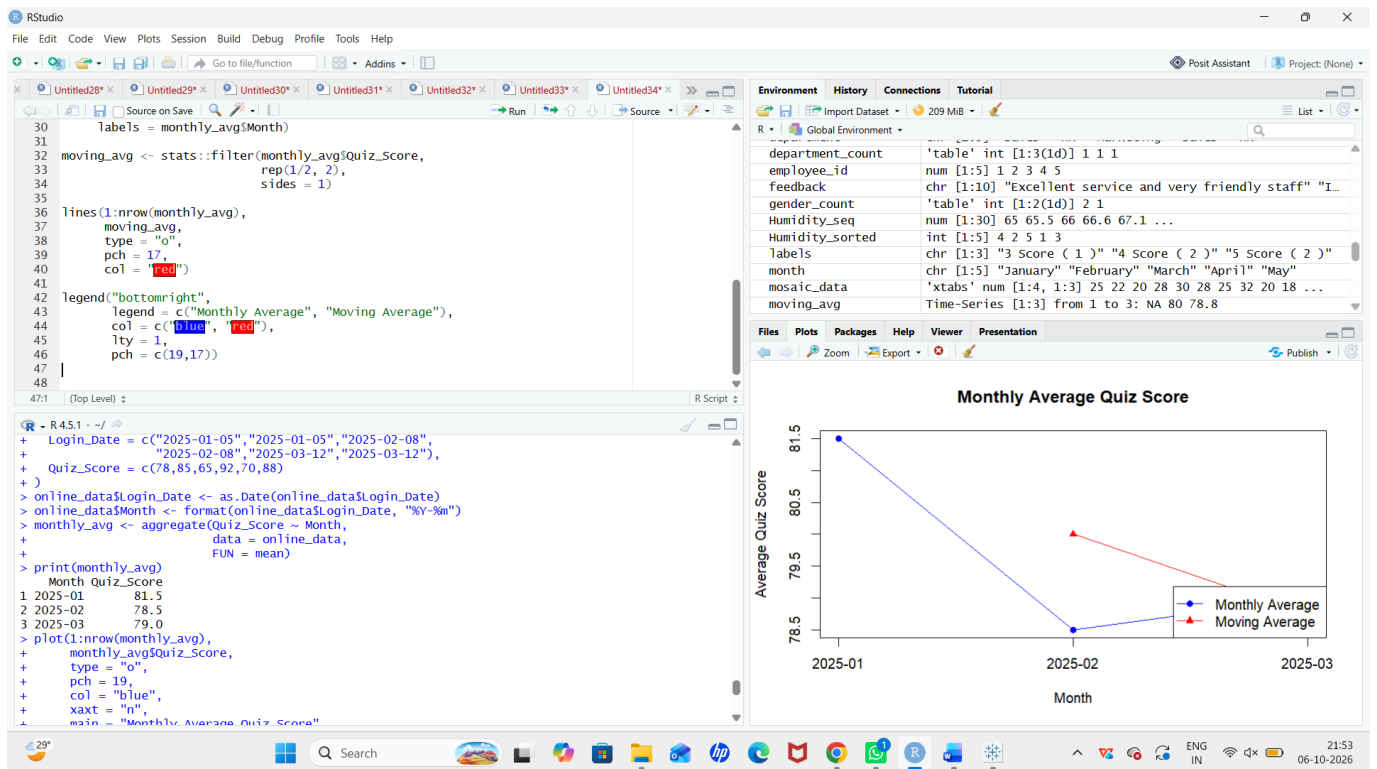
plot(1:nrow(monthly_avg),
     monthly_avg$Quiz_Score,
     type = "o",
     pch = 19,
     col = "blue",
     xaxt = "n",
     main = "Monthly Average Quiz Score",
     xlab = "Month",
     ylab = "Average Quiz Score")

axis(1,
     at = 1:nrow(monthly_avg),
     labels = monthly_avg$Month)

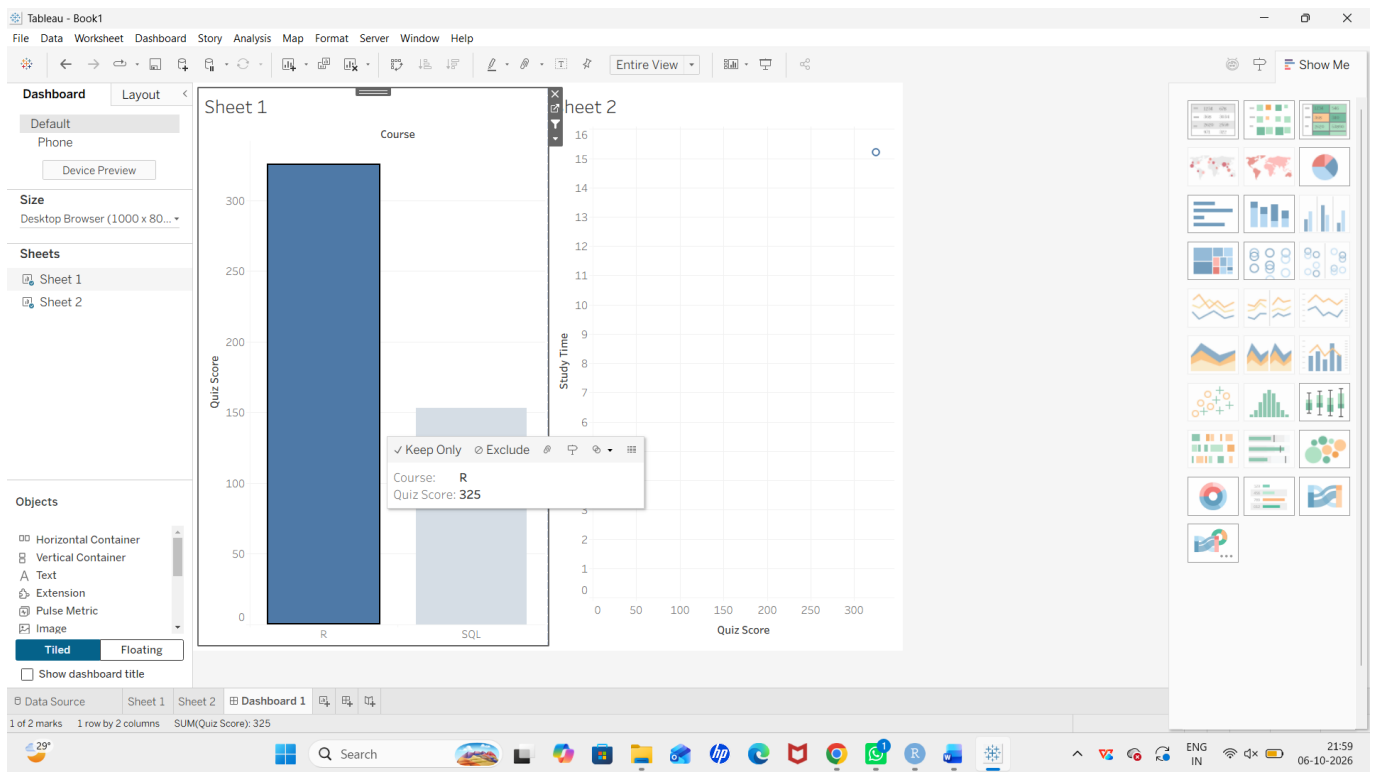
moving_avg <- stats::filter(monthly_avg$Quiz_Score,
                             rep(1/2, 2),
                             sides = 1)

lines(1:nrow(monthly_avg),
      moving_avg,
      type = "o",
      pch = 17,
      col = "red")

legend("bottomright",
      legend = c("Monthly Average", "Moving Average"),
      col = c("blue", "red"),
      lty = 1,
      pch = c(19, 17))
```



14.



PROCEDURE

1. Create and import the Online Learning Activity dataset into R.
2. Create a histogram to visualize the distribution of quiz scores.
3. Create a boxplot to compare quiz scores between courses.
4. Create a scatter plot between study time and quiz score, using videos watched as bubble size.
5. Convert the login date into Date format.
6. Calculate the average quiz score for each month.
7. Create a line chart with moving average smoothing to identify the performance trend.
8. Export the dataset as a CSV file for Tableau.
9. Import the CSV file into Tableau.
10. Create a bar chart showing average quiz score by course.
11. Create a scatter plot showing study time versus quiz score.
12. Add Gender and Course as filters.
13. Combine the charts to create an interactive Tableau dashboard.

RESULT

The online learning activity data was successfully analyzed using R. The histogram and boxplot showed quiz score distribution and course-wise performance. The scatter plot showed the relationship between study time and quiz scores. The monthly line chart with moving average helped identify the performance trend. Finally, an interactive Tableau dashboard was created using course-wise quiz scores and study activity.

